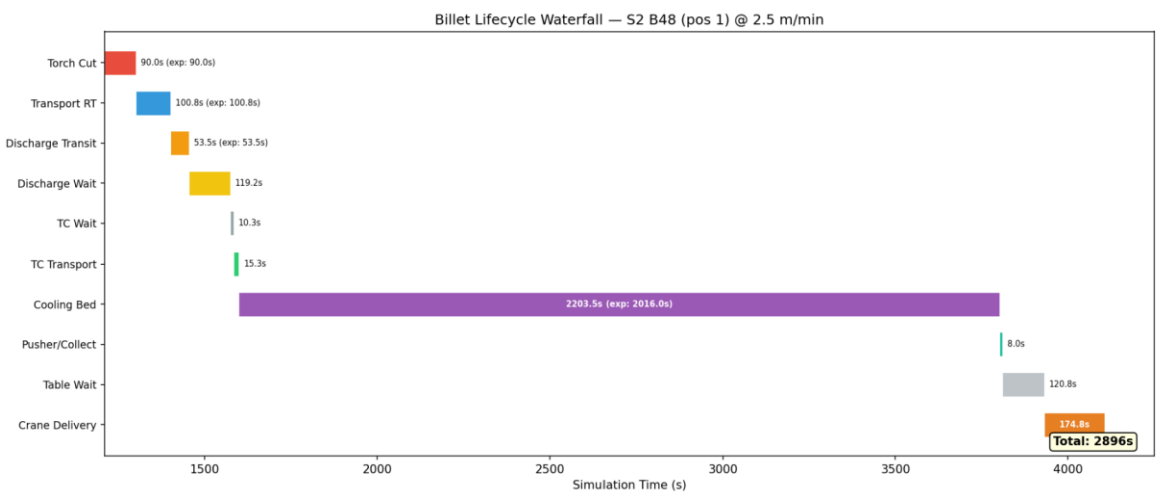
Velocity sweep experiments across 50 random seeds identify 2.7 m/min as the maximum velocity that produces zero traffic jams.

Comment1: what does this mean? Explain more

Discharge roller table. The billet continues at 15 m/min along a 13.375 m table (53.5 s). A movable stopper, located 6.2 m from the fixed stopper, positions the billet pair for pickup. A collision check at the movable stopper detects traffic jams after 28.7 s of transit

Comment2: how did this happen? Explain fully how 28.7s can be hand calculated.



Comment3: hand calculate this bar chart and show a bar chart with a couple of billets not only one. Like the refrence document I gave you.

Equipment Utilisation

Figure 2 shows the equipment utilisation breakdown at 2.5 m/min. All equipment operates in the green zone (below 70% utilisation). The transfer car is the most loaded resource at 56.5%, followed by the collecting table at approximately 31%. The cooling bed, cranes, and roller tables all operate with substantial headroom

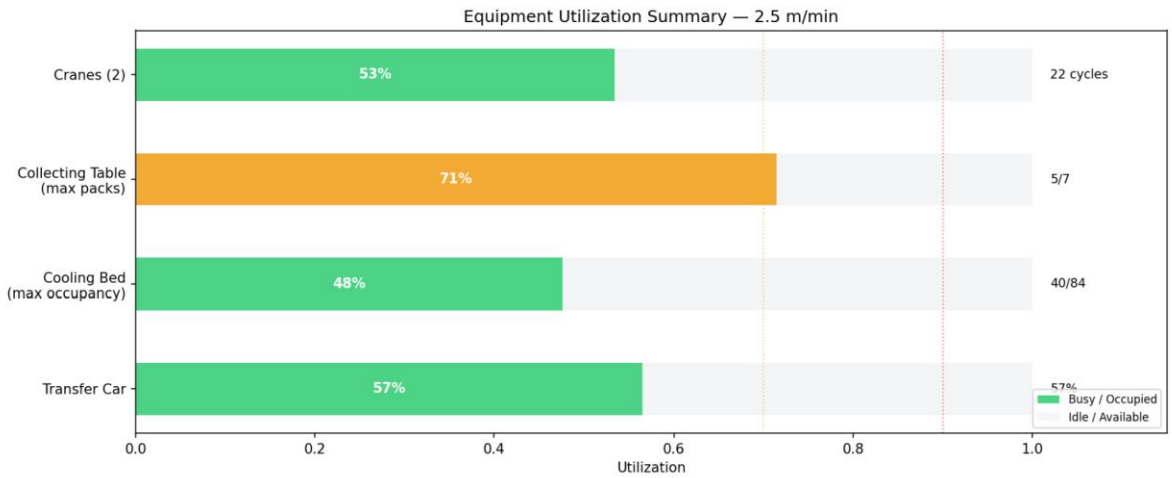


Figure 2: Equipment utilisation at 2.5 m/min. All components remain in the green zone ($< 70\%$). The transfer car is the most utilised resource at 56.5%.

Comment4: what is 71% that we see in the figure 2 and what is 31% that is stated int he text?

Cooling Bed Behaviour

Figure 4 shows the cooling bed occupancy heatmap. The characteristic diagonal bands confirm correct walking-beam operation: billets enter at slot 1 and advance one slot per 24 s cycle. Peak occupancy reaches 40 slots (48% of the 84-slot capacity), and mean occupancy is 29.8 slots, indicating that the cooling bed is far from being a bottleneck at this velocity

Comment5: I didn't get to know what occupancy is 29.8 slot means

Failure Modes

Two distinct traffic jam mechanisms are observed:

1. **Discharge roller table collision.** A new billet arrives at the movable stopper while the previous pair is still waiting for the transfer car. The collision is detected after 28.7 s of discharge transit (at the stopper position), not at the table entrance, giving the transfer car extra margin.
2. **Collecting table overflow.** When the table accumulates more than 7 packs before a crane can clear them, subsequent billets from the cooling bed have nowhere to go. This mode is secondary and typically follows a discharge jam

Comment6: didn't understand this!

Wait Time Degradation

Comment7: I have this comment generally and I am saying this on this one. you have to explain what your title is. for example, what is wait time degradation. and fully explain how you extract these numbers in the text so i can verify it. i don't know that your numbers are right or not. how should i verify these?

Cooling Bed Densification

Figure 9 shows the cooling bed heatmap at 4.0 m/min. The diagonal bands are denser, and peak occupancy rises from 40 to 60 slots (48% to 71%). While the cooling bed itself is not the bottleneck, the increased density reflects the higher pipeline pressure propagating downstream from the overloaded transfer car.

Comment8: is this a metaphor? please don't use this in the context of engineering reports!

Bug 3: Crane Throughput Bottleneck (Critical — Every Velocity Jams)

Symptom: Every velocity (even 2.0 m/min) resulted in a traffic jam at the collecting table.

Root cause: The initial crane model picked up 1 pack (2 billets) per trip, with a 350 s worst-case cycle. With 2 cranes:

- Max crane throughput: 1 pack per 175 s
- At 2.0 m/min, 6 strands produce pairs every ⍰180 s each → total ⍰1 pair per 30 s → 1 pack per 60 s
- Cranes: 1 pack/175 s, Demand: 1 pack/60 s → cranes can only sustain 0.69 m/min

This made the crane the bottleneck — collecting table overflowed at every velocity.

Fix: Changed crane model to bundle pickup: CRANE PACKS PER TRIP = TABLE CAPACITY (7 packs per trip). This is physically accurate because:

- Net crane capacity: 27 tonnes
- 7 × 2 × 0.8 t ≈ 11.2 t ⍰ 27 t
- Real plants use bundle grabs, not individual pack picks.

Comment9: you just fixed this theoretically but our crane only can pick up pack of two billets in each trip. and the problem still exists. you raised a good point in theory even with 2m/min velocity the crane still is a bottle neck. instead of that tell me instead of 6 strands 5 strands shall work or less. or the velocity shall be deduced or the crane has to lift pack of three billets. these are the only solutions you could give me. because my crane is not magnet its grab type and it is risky to lift too many billets at once.